

How AI Can Help Learn Lessons from Incident Reporting Systems

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Abstract—Both public agencies and private industry are learning from prior incidents using incident reporting systems to improve safety performance and reduce accidents. Incident reports are collected on anomalies of varying magnitudes—usually heavily weighted towards smaller incidents since they are more common than large ones. These reporting systems that collect information on anomalies are widespread across transportation, manufacturing, energy, and healthcare and are regarded as a foundational quality and safety improvement activity to which organizations are devoting considerable resources. For example, for decades, airlines and the Federal Aviation Administration (FAA) have placed a significant emphasis on voluntary data-gathering programs that enable airlines and government regulatory agencies to spot and correct problems before they lead to accidents. One such data collection system, the Aviation Safety Reporting System (ASRS) administered by the National Aeronautics and Space Administration (NASA), first established in 1976, receives more than 50,000 voluntary reports of safety incidents and situations each year from pilots, air traffic controllers, cabin crew, dispatchers, and maintenance technicians. Analyzing data from incident reporting systems can help an organization identify emerging risks, and this approach has been advocated for decades to prevent larger failures, but trends are difficult to find when there are more than 50,000 reports per year. This paper explores how AI tools can be used to find important lessons in incident reporting systems from near-miss events.

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1. INTRODUCTION

In the era of advanced technology and data-driven decision-making, two trends are emerging: 1) the recognition that near-miss data can be important predictors of future worse events and 2) enhanced artificial intelligence (AI) and machine learning (ML) tools that make big data analytics easily accessible for many organizations. This paper explores the intersection of AI and ML tools and aviation near-miss data, exploring the pivotal role that these tools can play in identifying near misses in data from safety reporting systems when key details of the event are captured primarily in narrative text descriptions of the event. Near misses, often subtle events with no clear warnings that something bad nearly happened, hold immense significance in enhancing safety measures and preventing catastrophic events. Harnessing the power of AI and ML to sift through vast volumes of data, recognize patterns, and discern the nuanced differences between near misses and routine incidents in text descriptions using Natural Language Processing (NLP) has the potential to further improve aviation safety. This paper investigates the challenges and implications of employing AI/ML/NLP in this critical context.

2. LEARNING FROM NEAR MISSES

Hindsight after a major accident provides a clear 20/20 vision, yet before disasters occur, how easy is it to recognize near-miss warnings? On March 5, 2000, Southwest Flight 1455 crashed through a fourteen-foot-high metal blast fence at the departure end of Runway 8 at Burbank, CA, continuing past the airport boundary, crossing a street, and coming to a stop near a gas station. There were 142 people on-board; 2 were seriously injured, 41 passengers plus the captain received minor injuries, and the airplane was extensively damaged [1]. After serious accidents, federal agencies and airlines closely examine events to learn from the mistakes [2,3]. Although searching for solutions and instituting new organizational routines is costly, failure motivates organizations and leaders to be open to change, to deploy resources to search for solutions, and to adopt new practices. Alternatively, experience with successes commonly stabilizes organizational practices and routines. Therefore, how individuals and organizations interpret near misses is critical for the learning that is likely to occur. If near misses

are interpreted as evidence of systemic success then leaders often fail to learn from these low-cost warning signals.

In the case of the Southwest Airlines-Burbank flight, they traced the crash to the instability of the approach that required a descent that was too steep and required too much velocity. Most air carriers, following this accident, developed procedures and trained crews to avoid unstable approaches. Moreover, if such approaches occurred, crews were trained to respond by executing a missed approach [4]. Nevertheless, the steep and rapid approach undertaken by Flight 1455 was a generally accepted practice for flight operations at many airports including at Burbank (although there is no available data from that time to ascertain its frequency specifically at Burbank). Investigators found that the pilots of Flight 1455 thought they had the situation under control up until touchdown because “they were using the same techniques to control their descent profile that they had used on previous ‘slam-dunk’ approaches which all worked successfully” [1, p. 68]. The first officer recounted to investigators that he had observed instances where captains successfully executed landings “when the airplane had not been fully configured at 1,000 feet, when it had been ‘out of the slot’ at 500 feet, and when idle thrust was required on final approach” [1, p. 75]. In interviews conducted with Burbank air traffic controllers following the accident, they shared their experiences of having “witnessed airliners ‘make it’ down to a landing on Runway 8 from altitudes as high as 7,000 feet or at speeds as fast as 260 knots ground speed over [the] Van Nuys [navigational aid]” [1, p. 76]. Previous incidents involving near misses frequently reinforce risky behaviors by fostering the perception that such behaviors are resilient to potential hazards. This accident happened in Burbank in particular because Runway 8 at Burbank is relatively short for a Boeing 737 (6,032 feet of paved surface), and investigators believe that had the runway been longer, the “accident would probably not have happened even if everything else was the same” [1, p. 78]. Had the runway been longer and no accident occurred, it is unlikely that the resulting near miss would have triggered the procedural change. Yet, for airlines to continue to improve safety, the industry needs to attend to the yet undiscovered or unrecognized risks in the system without waiting for an accident to bring attention to them.

Recent research [5, 6, 7] studied different types of “near misses,” and focused on near-miss events where a negative outcome could have happened because of hazardous conditions but did not because of chance [5]. Tinsley et al. [7] demonstrate that some of these near misses have cues that highlight resilience (i.e., a disaster was successfully avoided – a “resilient near miss”), while others highlight vulnerability (i.e., a disaster nearly occurred – a “vulnerable near miss”). In a series of studies, they show that when near misses have salient information about danger, the events will prompt people to scrutinize near misses more critically. In contrast, for events without this clear danger cue, people will confidently view these near-miss events as successes, fail to learn from the event, and thus fail to prepare for the next potential hazard. Therefore, it is critical to first capture all

possible near-miss events in incident reporting systems including what went right and what went wrong, and to identify which types of near misses are occurring to focus on learning the correct lessons [8].

3. INCIDENT REPORTING SYSTEMS

Near misses improve safety because they commonly occur at higher rates than accidents [9, 10], so they provide low consequence insights into safety prevention, and because they are successful outcomes, they provide information on a system’s capacity for resilience [11]. Incident or anomaly reporting systems are important to collect this incident data for learning by encouraging (and sometimes requiring) the reporting of occurrences to a central entity that collects and then uses the data for safety prevention purposes. Many industry-wide systems exist across transportation, manufacturing, energy, and healthcare [12], and it is important for current reporting systems to capture appropriate information to understand near misses.

Many industries are interested in learning from near misses and are collecting incident reports, and most transportation industries have seen decreases in accidents over time. This section describes several datasets from the aviation industry that aggregate incident data.

Three data sets of publicly available information are available for incidents in the U.S. commercial aviation industry. First, the Aviation Safety Reporting System (ASRS), operated by NASA and the Federal Aviation Administration (FAA), which is also a USDOT agency, provides information on near misses and also some accidents and incidents (see Table 1). The system relies on voluntarily submitted reports from pilots, controllers, flight attendants, mechanics, ground personnel, and others. The reports are reviewed by analysts who anonymize submitted reports, provide labels for structured fields and categories, and since 2009 have coded human factors information. Reports are kept confidential, and reporters are given immunity from enforcement actions in most cases. The information collected and maintained by the ASRS is highly subjective based on the voluntary nature of the system and how the data are collected, but ASRS is generally considered to be an exemplar for near-miss reporting systems and data from ASRS have been used in published studies on near misses.

The second system is the National Transportation Safety Board (NTSB) Aviation Accident Database. This database documents all aviation accidents as well as aviation incidents that NTSB investigates. The system captures all incidents the NTSB examines (Table 2). The reporting of incidents to the system is discretionary only in the fact that after an initial notification, the NTSB will decide whether to investigate or not. This results in some subjectivity inherent in the decision to investigate.

The third system for near-miss incidents in the U.S. commercial aviation industry is the Accident and Incident

Data System (AIDS), maintained by the FAA (Table 3). Reports in AIDS describe aviation accidents and incidents that have an impact on aircraft safety, but do not meet the threshold for an accident as defined by the National Transportation Safety Board (NTSB). The system has clear reporting requirements and all accidents, as well as incidents

meeting certain criteria (as outlined in FAA order 8020-11C), must be reported. For incidents that fall below the given criteria (including near misses) reporting is voluntary in nature. For accidents meeting the reporting criteria, FAA can bring legal action on individuals for failure to report an accident or reportable incident to the system.

Table 1: Details of ASRS

System	Aviation Safety Reporting System (ASRS)
Agency	Operated by the National Aeronautics and Space Administration (NASA) for the Federal Aviation Administration (FAA)
Website	https://asrs.arc.nasa.gov/
What is the time frame?	Data since 1988 is available from the website
What events are reported?	"Voluntarily submitted aviation safety incident/situation reports from pilots, controllers, and others." ASRS is primarily designed to collect information on near misses, but it also may contain reports relating to accidents and incidents.
How clear are reporting requirements?	Subject to interpretation by reporters. "Pilots, air traffic controllers, flight attendants, mechanics, ground personnel, and others involved in aviation operations submit reports to the ASRS when they are involved in, or observe, an incident or situation in which aviation safety may have been compromised."
How much discretion in what is reported?	A great deal. All reports are voluntary. What constitutes a situation where "safety may have been compromised" is subjective. Needs to report if there is a violation of a Federal Aviation Regulation (FAR).
Is reporting required or voluntary?	Voluntary. Identities of reporters are kept confidential, and reporters are given immunity from enforcement actions in most cases.
Who reports events?	Anyone involved in commercial aviation may report. Primarily pilots, air traffic controllers, flight attendants, mechanics, and ground personnel.
How are reports collected?	Reports may be made electronically through the ASRS website, or mailed in in paper format.
Does a 3rd party aggregate or scrub the data?	Yes. Reports are received, processed, and de-identified by NASA.
What is the volume of reports annually?	Over 50,000 reports annually
Are data released publicly for the system?	Yes. De-identified reports are made available publicly.
What are the protections against discovery or future lawsuits?	If a reported incident involves a policy violation by the reporter, the reporter is given immunity from enforcement action except where the violation was deliberate or criminal, or where the reporter has a history of enforcement actions.
Other important links	https://asrs.arc.nasa.gov/search/database.html
	https://asrs.arc.nasa.gov/publications/callback.html

Table 2: Details of NTSB Aviation Accident Database

System	Aviation Accident Database
Agency	National Transportation Safety Board (NTSB)
Website	https://data.nts.gov/avdata
What is the time frame?	Data available for online search from 1962 to present
What events are reported?	Aviation accidents and aviation incidents that NTSB examines
How clear are reporting requirements?	Clear. Data set captures all incidents NTSB examines
How much discretion in what is reported?	Discretion occurs in that after an initial notification of an incident, NTSB decides whether to investigate or not.
Is reporting required or voluntary?	This dataset is capturing only the incidents that NTSB investigates.
Who reports events?	NTSB investigators and specialists gather/document the investigation.
How are reports collected?	Once the final accident investigation report has been adopted, the final copy of the report will be prepared for public release.
Does a 3rd party aggregate or scrub the data?	No, but reports need to be approved by Board Members.
What is the volume of reports annually?	About 1,200 aviation accidents and incidents a year.
Are data released publicly for the system?	Yes. Reports are publicly available.
What are the protections against discovery or future lawsuits?	None. NTSB is attempting to identify probable cause.
Other important links	https://www.nts.gov/investigations/process/Pages/default.aspx

Table 3: FAA AIDS

System	FAA Accident and Incident Data System (AIDS)
Agency	Federal Aviation Administration (FAA)
Website	https://www.asias.faa.gov/apex/f?p=100:12:::NO::
What is the time frame?	Data available for online search from 1995 to present
What events are reported?	AIDS contains reports of aviation accidents and aviation incidents, which are defined as events that affect the safety of aircraft operations but does not meet the threshold to be defined by the NTSB as an accident.
How clear are reporting requirements?	Clear. All accidents must be reported. Incidents meeting certain criteria must also be reported. FAA order 8020-11C provides detailed guidance on reportable incidents. Other, smaller incidents may also be reported. Incidents in which no significant injuries or damage occurred may be considered near misses.
How much discretion in what is reported?	Very little. Discretion is only exercised for small incidents below the FAA's reporting threshold.
Is reporting required or voluntary?	Required.
Who reports events?	Pilots, air traffic controllers, and flight crews.
How are reports collected?	Reporters fill out a paper form or report by phone (although a written follow-up may be requested).
Does a 3rd party aggregate or scrub the data?	No. Data are handled by the FAA.
What is the volume of reports annually?	About 1,000 incidents reported annually.
Are data released publicly for the system?	Yes. Reports can be accessed by the public.
What are the protections against discovery or future lawsuits?	Reporters may be prosecuted for failure to report an accident or reportable incident.
Other important links	https://www.asias.faa.gov/apex/f?p=100:15:::NO::P15_REGION_VAR:1

4. PAST EFFORTS TO USE AI/ML/NLP TO ANALYZE INCIDENT DATA SETS

When incident data are publicly available, these data sets often have been used in proof-of-concept AI/ML/NLP models because of the volume and complexity of the data set. For example, Kong et al. [13] examined data regarding road traffic accidents and Gadekar & Bugalia [14] and Bugalia et al. [15] examined data from construction site incidents. Here we focus on those using commercial aviation data sets since the ASRS dataset in particular has been a sample data set used in many proof of concept AI/ML/NLP models over the last 15 years [16].

In 2005, Posse et al. [16] focused on aviation anecdotal reports from the ASRS system which at the time had more than 600,000 narratives. Given the potential wealth of information locked up in the database in anecdotal narrative text fields, Posse et al. [16] proposed a methodology and algorithm for automatically extracting human performance factors from the ASRS safety narratives. Their work analyzed twenty ASRS reports with their “regular expression analysis” [16, p. 2] and compared these results with four human subjects that read the reports and identified human performance factors. They demonstrated that their algorithm (GATE) was very much in agreement with two of the human scorers but less with the other two (which also shows the human scores may not always be consistent). This early approach was limited by a heavy need for domain-specific knowledge from experts, but the fourteen factors that they identified in their study were used in many more projects (for example, Persing and Ng [17] and Abedin et al. [18]) that attempted to use AI/ML/NLP algorithms to extract knowledge from the ASRS dataset.

Many attempts followed where researchers used an initial set of labels and a portion of the data as a training data set that included incident reports labeled by subject matter experts (for example, Wolfe [19] and Persing and Ng [18]). More recent attempts include a mix of supervised learning relying on prior classifications and unstructured learning that does not require any domain experts.

Tanguy et al. [20] describe multiple tools that analyze the ASRS dataset and an equivalent European data. First, they use NLP techniques to infer some of the metadata descriptors from the text narratives to create automatic report classifications. This task relied on supervised machine learning algorithms (Support Vector Machines) constructed from a volume of previously categorized data. The model worked well for several categories when those categories frequently appeared in the dataset and commonly caused the same problem (i.e., bird strikes). The model worked less well when the initiating event was the failure of a non-engine system component which could lead to many different possible problem outcomes. It also worked less well for categories that appeared rarely in the dataset. Second, they

describe the use of probabilistic topic modeling which assumes that documents are a “bag” of words, that documents include topics of varying importance according to a specific distribution, and topics are expressed with words according to a specific distribution. Thus, one can estimate these distributions from observed frequencies of words in documents. While they felt this modeling was generally successful, the challenge was the granularity of the extracted topics compared to any indexing of the reports. Basically, this approach confirmed the categorizations that were already in the data and did not provide any new insights from the data. For a different dataset without prior metadata (i.e., categories), this could be a valuable technique. Third, they describe a system in use by aviation safety experts in France that will take a source report and will identify similar reports among those indexed in the database based only on the narratives in the report. Tanguy et al. [20] describe how they calculated the similarity score “based on the lexical overlap of the narrative parts of two documents... The more words they share in common the more similar the documents are” [20, p. 90]. Fourth, they describe a variant of machine learning called active learning that is based on the traditional supervised learning methods but only submits items designated as borderline to an expert for judging. They use this approach for creating a new dimension or new metadata attribute without requiring extensive coding of all existing reports. The domain experts only focus on the borderline cases.

Kuhn [21] describes using structural topic modeling (STM), an unsupervised learning method, to analyze the ASRS dataset which by their study included 1.3 million records. Their results were similar to those reported by Tanguy et al. [20] that this approach confirmed categorization already available in the ASRS dataset. They concluded that their method would be helpful at either a more detailed level (i.e., they provide a case study focused on incidents at only San Francisco Airport) or when trying to provide missing metadata or introducing a new dimension.

Dong et al. [22] used 200,000 incident reports from ASRS and a deep recurrent neural network model to identify primary and contributing factors in each incident report in an attempt to ease the burden on the analysts who currently review incident reports before they are added to the ASRS database. They focused on incident reports involving the six most frequent categories. Similarly, their models worked in that they confirmed the category data already in the dataset, and they concluded that their models would be able to identify categorical data for future reports if ASRS wanted to go away from analysts labeling incident factors.

Zhang & Mahadevan [23] is most relevant to our task because they describe a hybrid model consisting of a support vector machine based on a bag-of-words approach applied to the event synopsis and an ensemble of deep neural networks trained using categorical items to quantify the risks

associated with consequences in ASRS reports. They use the event synopsis to map event outcomes to risk levels based on a risk-based outcome categorization. If events have multiple consequences, they mapped it to the highest risk group for outcomes of that group. The models work well at identifying the worst consequence and then applying their algorithm to assign a risk level, again based on the outcome of the incident. We argue that the shortcoming of this approach is that it traps analysts in the outcome bias. The outcome bias is a well-known cognitive bias [24] that refers to the tendency to judge something based on the outcome instead of all the factors relevant to the situation. For example, most near misses have good outcomes but that does not mean they all have low risk. In this project, we seek to use event synopses to map events to risk levels, but rather than focusing only on the outcome, we need to explore the near-miss characteristics of each event.

Additionally, we have had prior success learning from near misses in incident reporting systems not in an aviation context but in maritime safety. Two of the authors used the US Coast Guard's incident reporting system called the Marine Information for Safety and Law Enforcement (MISLE) database to create a prediction model for future accidents [25]. Similar to the aviation datasets, accidents are rare events and minor incidents are widespread in this maritime data set. Thus, this project initially focused on determining if near-miss events identified in MISLE could be valuable predictors of worse future events. The analysis showed that recent near-miss counts in MISLE for the same vessel type and for other vessel types are useful for predicting future serious casualties at a waterway level. Using this finding, a predictive AI/ML model was built using distinct models for each waterway by vessel combination. Random forest decision tree AI/ML models were used to identify waterways of significant risk based on future total casualty costs for a waterway link as well as whether or not casualty costs are predicted to exceed certain predefined monetary thresholds in the next month. An R-based predictive model was designed to be run monthly, using data from prior months to make predictions for the subsequent month. The prediction models were tested on ten months of data from 2022. The overall accuracy of the predictions made by these models was 92%-99% depending on model characteristics. The predictions of the models were considered accurate enough to be potentially useful in future accident prevention efforts for the US Coast Guard and were considered generalizable and thus valuable to other industries that have near-miss data and a desire to increase future safety. We hope to extend this work to aviation datasets.

5. RECOMMENDATIONS FOR HOW AI/ML/NLP CAN HELP WITH NEAR MISSES

As more incidents occur over time in combination with an increased focus on the importance of reporting incidents, the volume of data on incidents is ever-increasing. Traditional methods of manual data collection and analysis already struggle to handle and analyze the sheer volume of incidents

being gathered, but AI and ML tools can help. AI has the capability to process large amounts of data extremely fast, allowing accurate analysis of incidents to be made quickly. ML models can be used to train existing data and be implemented to be able to recognize patterns in data, leading to a deeper understanding of the near-miss incidents and potentially coming up with a general model for them.

NLP for Near Miss Identification

Natural Language Processing (NLP) is a subset of AI that focuses specifically on computers being able to process and understand human languages. NLP is beneficial because so many of the important details of an incident are captured in the text narratives. As demonstrated in the prior examples, NLP is being used to analyze incident reports by directly taking these reports and producing an output that categorizes/classifies the event. The speed at which NLPs could do this would greatly reduce the total amount of time needed to process and analyze each individual incident, while still extracting all the necessary information from each report. Instead of having a person or human team look at each incident event and categorize by hand, each incident could be run through a NLP algorithm that could recognize certain keywords that would allow it to categorize each event as a near miss or not. NLP has the potential to streamline and increase the speed of classification and analysis of these events. The challenge for this project is training the NLP to recognize near-miss events and to differentiate the near misses that have important lessons to learn from the inconsequential events.

Incident Pattern Recognition and Predictive Analysis

AI can also be used to identify patterns and trends in incident databases through pattern recognition and machine learning. Pattern recognition takes accumulated data (in our case from incident management systems) and analyzes the historical data to predict and notice trends in data. These pattern recognition models could predict areas of potential risk, which would alert the proper review authorities about areas that are significantly more at risk, helping these authorities to be aware and implement countermeasures. This type of risk prediction is what two of the authors did for the US Coast Guard using their incident reporting system data [25]. The challenge with applying this to aviation data is that we found that the best models examined the system at the lowest level. In the US Coast Guard case, this was a particular vessel type on a particular waterway. While this project may eventually get to this level of detail, we are still focused at the highest level on determining if near-miss events can be identified in aviation data sets separate from other types of incidents.

Categorization of Near-Miss Incidents using Large Language Models (LLMs)

The use of Large Language Models (LLMs) is another example of technology that may be useful to classify near-miss incidents. LLMs have the ability to take human text, process it, and then generate human-like text results. The

usage of LLMs similar to those of the GPT line of LLMs could be used to take raw incident data gathered from incident databases and help identify characteristics of near misses in incidents. Similar to NLP, this would eliminate requiring human analyst examination of individual incidents to even begin. Algorithms could generate output that categorizes, ranks, or scores the incidents to aid and guide human analyst labor.

AI, NLP, and LLMs hold significant potential for the sorting, analysis, and categorization of aviation incidents. As demonstrated in the prior applications from the last 15 years, many models can categorize outcomes of events matching the prior classifications accurately. Our challenge is: can they distinguish near misses from other less concerning incidents?

6. PILOT STUDIES

For our pilot studies, we used data from the NTSB Aviation Accident Database, examining incidents from 1990-2007. An analysis of the near misses in this dataset is described in Madsen et al. [26] which found using fixed effects Poisson regressions that airlines learn from accidents (their own and others), and from one category of near misses—those where the possible dangers are salient. That analysis showed that airlines do not improve following near-miss incidents when the focal event has no clear warnings of significant danger and concluded that while airlines need to and can learn from certain near misses, they were not learning from all near misses. Similar to the ASRS dataset previously described, this dataset has some categorical data (injuries, fatalities, and damage) and text narratives with detailed event descriptions. What was not provided in the dataset was a categorical description of the cause of the incident. We had a licensed, currently active commercial airline pilot from one of the major carriers with more than 20 years flying for the airline, read and label about 1,200 incidents for event categories. Also based on their experience we developed Table 4 to distinguish events that could have been worse (near misses) from other less concerning incidents. Using this logic, the 1,200 events in this dataset were labeled as accident, near miss, or minor event.

NLP

Using the 1,200 incidents that were already categorized as accidents, minor events, and near misses from the NTSB data set, we used a bag-of-words algorithm for further analysis. First, all “stop words” are removed that are too common to be helpful (i.e., a, an, the, of, because), then the bag-of-words model represents each narrative as a vector of word counts. This means that each document is represented as a set of word counts with each word represented as a dimension in the vector (i.e., aircraft: 6, taxiway:4, tower: 3, etc.). The model treats each word as a separate entity, without regard for grammar or word order hence the term: bag-of-words [27].

Table 4. - Guidance for Categorizing Events

Not near misses

All major accidents in the dataset (already bad)
Severe turbulence (already bad)
Less concerning incidents: health issues, personal injuries, fumes/odor/minor smoke, ground collisions (slow speed)

Near misses

Runway incursion/near high-speed ground collision
Didn't land/depart on intended runway
Terrain proximity alert
Near mid-air collision
Landing gear failures/brakes fire/tire bursts/steering failure on landing
Engine Failure - contained or uncontained
Fire/serious smoke
Aborted take-offs
Bird strikes
Mild turbulence

We randomly pulled out one third of the data to use as a test data set and trained the model on the other two thirds of the data. We ran a random forest classification model using the results to try to classify each event as an accident, a near miss, or a minor event restricting the model to only use the narrative text for each individual record. We repeated the random sampling and splitting of the dataset a total of ten times and then averaged the performance metrics. We then repeated that process for the case where the random forest model could use the other NTSB data categories in addition to the narratives. Table 5 summarizes the results of the two models.

Allowing the model to use the other NTSB categorical variables helped make the model more accurate—especially in identifying accidents. It also helped a small amount for the near misses and minor events. The model that used all of the NTSB categorical variables had an overall classification accuracy greater than 90% for all three categories. The model that could only use the text was not as good but was still pretty good with greater than 80% for all three categories.

We are continuing to explore more complex language models and will compare them to the bag of words algorithm, but this worked well for a preliminary test.

Table 5. Results from the Random Forest Classification Model

Random Forest model using only narrative text			
Metric	Accidents	Near Misses	Minor Events
Classification Accuracy	.807	.825	.835
Sensitivity	.760	.731	.478
Specificity	.726	.849	.966
Random Forest model using narrative text and NTSB categorized data			
Classification Accuracy	.985	.915	.909
Sensitivity	.978	.835	.819
Specificity	.978	.902	.976

LLMs

LLMs such as the ones that power the ChatGPT product have been trained on a wide range of text and have been adapted to a substantial amount of general knowledge. Given its training, we can submit details for events that we are interested in and ChatGPT can analyze the text to interpret descriptions of events and apply the criteria provided to make categorizations. Figure 1A and Figure 1B show the descriptions of accidents, minor events, and near misses provided by us to ChatGPT. Figure 2 shows the summary provided by ChatGPT describing its interpretation of the sample input we provided. Figure 3 shows the three events that we provided for it to categorize. Figure 4 shows its categorizations. For these three cases, it identified the events consistent with our expert. This was not always the case for some of the events that we tested. In particular, ChatGPT had difficulty distinguishing between minor and serious injuries and tended to over-categorize some events as accidents that the NTSB had labeled as incidents. Future work could involve doing industry or enterprise-specific training involving human feedback on output, termed Reinforcement Learning from Human Feedback (RLHF) [28]. The LLM that powers ChatGPT (usually GPT 3.5 or GPT 4.0) had a portion of its training done using RLHF. This could open the pathway to leveraging AI to give deeper insights into the broader scope of large datasets like these to train a classification model based on actual events that have already identified categories. That should make the model better at identifying events consistent with how experts interpret events.

It should also be noted that most LLMs are merely trying to guess what text comes next. Given a context and a user prompt, modern LLMs will simply try to calculate what might come next, based on other texts the models have seen before. It is a bit like using only autocorrect to draft a tweet, just an industrial-strength version. These models are not doing any calculations in the typical “calculator” sense, but are merely extremely good at, and extremely fast at, guessing the word that comes next. There is no tabulation or word-counts or document-term-frequencies used as in some more mechanical and deterministic methods, and the results are

often unexplainable in any causal sense. It is also important to emphasize that the model does not “understand” anything, it is merely guessing what word “sounds best” next in its answer based on history.

7. ETHICAL CONSIDERATIONS AND OTHER LIMITATIONS

AI and ML tools should be used with caution. There are always concerns regarding the datasets and also interpretations of the data by models. For example, only a subset of received ASRS reports are entered into the database by analysts. Modelers need to recognize if this could produce biases in results in some way. Also, the events are not independently verified and may represent only one side of a story. Additionally, there is an inherent imbalance of the types of events. This data distribution is likely to cause the classifiers to be biased toward the dominant categories or classes which in the case of ASRS reports are human factors causing the incident. This bias may reinforce Haunschild and Sullivan’s [29] finding that when events have serious consequences, there is a disposition for organizations to “blame the operator” [29, p. 614] and in aviation they blame the pilots. Aside from all this is the occurrence of “hallucinations” of LLMs inherent to their nature as a content generating device. Zhang et al. describe how LLMs “occasionally generate content that diverges from the user input, contradicts previously generated context, or misaligned with established world knowledge” [23].

Regarding interpreting results, reporters from the New York Times examined the ASRS dataset and found: “In the most recent 12-month period for which data were available, there were about 300 accounts of near collisions involving commercial airlines” [30]. This was described as “part of an alarming pattern of safety lapses and near misses in the skies and on the runways of the United States, a Times investigation found.” None of which resulted in an accident, and the article focused on Conflict Alerts and how many are in the dataset. But Freidman-Berg et al. [31] estimated that 62% of CAs in the enroute environment and 44% in the terminal environment did not require intervention by a controller. For example, there can be a correctly operating CA when two aircraft are approaching head-on if they are on converging approaches to parallel runways and will continue to be safely separated while remaining on their approach paths. Alternatively, two aircraft on curved approaches to parallel runways will generate a CA as the aircraft approach head-on since a collision would occur if they continued on their trajectories, but instead they will continue turning to their parallel inbound final approach paths. As AI and ML tools make accessing the datasets and findings easy, care must be taken for what conclusions are drawn from the data.

Ensuring that sensitive information is handled properly, recognizing potential biases, and being transparent about the use of AI and ML technologies are paramount to maintaining public trust in all the partners working to improve aviation safety.



Studying prior incidents is important because it can lead to important safety improvements that reduce future risks. However, not all incidents have the same importance. Accidents are the easiest to distinguish because there are significant aircraft damage and serious injuries or fatalities. The remaining incidents would have only minor aircraft damage, minor injuries or no aircraft damage or no injuries. Of the incidents that are not accidents, we would like to determine which incidents were near misses. A near miss is an incident where if not for some good luck, the incident could have been worse. The element that distinguishes a near miss is the presence of a much worse consequence that was prevented only by good luck.

For example, this event is an accident:

A CRJ200 had completed the pushback process and was waiting for taxi clearance. A CRJ700 was being pushed back from another gate, with a wing walker stationed at the left wing. The wing walker was aware of the CRJ200, and when the pushback commenced, he believed that the tugdriver was only going to pushback the airplane about 10 feet. As the tugdriver continued to push the airplane past 10 feet, the wing walker signaled the tugdriver to stop. Witnesses observed the wing walker signaling the tugdriver to stop as the tugdriver continued to push the airplane. The tail section of the CRJ700 struck the tail section of the CRJ200, which resulted in substantial damage to the empennages of both airplanes. Due to lightning in the vicinity of the airport, ground crewmembers were not using headsets.

This is an example of a minor event that is not a near miss. The damage was not substantial and good luck did not prevent the event from being worse:

The airplane was on the south ramp, taxiing from landing, when the right wingtip struck a blast fence. The pilot reported he was following instructions of the ground marshallers at the time. He reported that he was taxiing at about 4 knots, when the marshaller signaled him to turn left. The pilot made a hard left turn, and three quarters into the turn, he felt the nosewheel jitter. He looked out the window in front of the airplane and noticed the marshal signaling him to stop.

Figure 1A - Input Text Provided to ChatGPT to Explain the Incident Types

8. CONCLUSIONS

While still in the pilot stage of data analysis, this project has demonstrated that AI and NLP algorithms have the potential to help identify near misses in incident data sets. But even though AI and NLP can significantly improve efficiency in incident analysis and categorization, it is crucial to understand that these technologies are not meant to replace human experts. Instead, they serve as valuable tools that augment human decision-making processes. Human analysts still need to validate and interpret the AI-generated results, ensuring the accuracy and relevance of the final assessments.

of learning from past experiences and continuously improving operations. One of the ways in which NASA documents and shares lessons learned is through its Lessons Learned Information System (LLIS). From reviewing publicly available data sources, it seems that interest in a centralized knowledge repository for NASA lessons learned has lagged in recent years. Demonstrating the value of capturing anomalies and incident data and the benefits of AI/ML tools to analyze stored data can hopefully revitalize this mechanism for teams to learn from past experiences and apply those lessons to future endeavors.

Finally, NASA and commercial aviation have strong cultures

This is an example of a minor event that is a near miss: Aircraft experienced collapse of the left main landing gear bogie while stopped at the runway awaiting takeoff clearance. Post-incident examination indicated the aft bogie swivel joint lugs has failed due to stress corrosion cracking. The swivel joint did not have proper lubrication which keeps water out. This allowed formation of corrosion on the surface of the lugs, and subsequent crack initiation at the corrosion pits. Operator maintenance requirements specified that the swivel joint be lubricated every weekend check and every 'A' check. The work cards for these two inspections called out in narrative form the lubricating of the bogie pivot joint. The cards called for the swivel joint to be lubricated by reference to a number on an illustration. The failure of the swivel joint lugs was the subject of Douglas Aircraft all operator letters in 1976 and 1990. These letters stressed to operators the importance of lubricating the swivel joint to prevent corrosion formation.

It was good luck that the problem with the landing gear happened while the airplane was stopped and not during takeoff or landing.

In general, we believe that incidents that did not have substantial damage or substantial injuries or fatalities that involved runway incursions, high speed ground collisions, near impact with terrain, not departing or landing on intended runway, near mid-air collisions, landing gear failures, brake fires, tire bursts, steering failures, engine failures, fires, serious smoke. Aborted takeoffs, bird strikes, and mild turbulence are all near misses. We believe that incidents that did not have substantial damage or substantial injuries or fatalities that are minor events and not near misses are slow speed ground collisions, health issues, personal injuries, minor smoke, fumes or odors.

Figure 1B - Continuation of Input Text Provided to ChatGPT



Thank you for providing clear examples and criteria for distinguishing between accidents, near misses, and minor events in the context of aviation incidents. Your explanation helps highlight the importance of studying these incidents for safety improvements. Here's a summary of your distinctions:

1. Accidents:

- Significant aircraft damage.
- Serious injuries or fatalities.

2. Near Misses:

- Incidents where a much worse consequence was prevented only by good luck.
- Examples include runway incursions, high-speed ground collisions, near impact with terrain, not departing or landing on the intended runway, near mid-air collisions, landing gear failures, brake fires, tire bursts, steering failures, engine failures, fires, serious smoke, aborted takeoffs, bird strikes, and mild turbulence.
- These incidents didn't result in substantial damage or substantial injuries/fatalities.

3. Minor Events (Not Near Misses):

- Incidents that are not near misses but are also not accidents.
- Examples include slow-speed ground collisions, health issues, personal injuries, minor smoke, fumes, or odors.
- These incidents didn't result in substantial damage or substantial injuries/fatalities.

Your criteria provide a useful framework for categorizing aviation incidents and prioritizing safety improvements based on their potential consequences. Studying near misses, in particular, can be valuable for identifying weaknesses in safety procedures and implementing corrective measures before a more serious accident occurs.

Figure 2 - ChatGPT's summary of the information provided

- 1) On August 4, 2009, a Pinnacle Airlines, Inc., CRJ-200, operating as flight number 2871, was enroute from Knoxville, TN to Detroit, MI when the flight encountered turbulence. The flight had been operating at FL300 and had just initiated a climb over cumulonimbus clouds when the turbulence was encountered. The seat belt sign was on and the flight attendant was conducting the in-flight service. The flight diverted to Louisville, KY and landed uneventfully. There were 24 passengers on board and 3 crew. The one flight attendant received a serious injury and one passenger received a minor injury. The flight attendant was transported to a local hospital and the passenger was treated at the airport and released.
- 2) N6815 was pushed back from Gate H-3 into the gate alley. As the airplane was being turned out to taxi for takeoff, the horizontal stabilized contacted the elevator of another Boeing 727 which was parked at Gate H-1. The flight crew of N6815 had received a release salute from the ground crew prior to the taxi.
- 3) FLT 1206 was on final for a Cat II ILS RWY 2L APCH. The last reported RVR prior to landing was 700 ft, & 1200 ft RVR was reported as the flight crossed the outer marker. Minimums for the approach were 1200 FT RVR on short final. The capt saw approach lights and disconnected the autopilot as the aircraft descended below 200 FT AGL. The aircraft touched down 408 ft short of the runway threshold, damaging the approach lighting system and the aircraft. DFDR readout revealed the aircraft rate of descent increased from 720 FPM to 1400 FPM after the autopilot was disconnected, while airspeed remained constant. Inspection of the aircraft nav equipment and airport nav aids revealed no evidence of malfunction. Although the capt stated that he had the approach lights visually, he was unaware the aircraft had landed short. The capt disconnected the autopilot at about 200 feed AGL, although company procedures required use of the autopilot down to decision height (100 FT AGL).

Figure 3 - Events provided to ChatGPT to Categorize



Certainly, here's the categorization of the three events into accidents, near misses, and minor events (not near misses).

1. Event 1:

- Description: A CRJ-200 enroute from Knoxville to Detroit encountered turbulence during a flight. The flight diverted to Louisville, KY and landed uneventfully. There were 24 passengers and 3 crew onboard, with one flight attendant receiving a serious injury and one passenger receiving a minor injury.
- Categorization: Accident
- Reasoning: This incident results in a serious injury to a flight attendant and a minor injury to a passenger. Therefore, it qualifies as an accident.

2. Event 2:

- Description: Aircraft N6815 was pushed back from gate H-3 into the gate alley. During the taxi for takeoff, the horizontal stabilize of N6815 contacted the elevator of another Boeing 727 parked at gate H-1. The flight crew had received a release salute from the ground crew prior to taxi.
- Categorization: Minor Event (Not a Near Miss)
- Reasoning: This event involved an aircraft contact but did not result in substantial injuries or significant damage. Therefore, it is classified as a minor event, not a near miss.

3. Event 3:

- Description: Flight 1206 was on final for a Category II ILS RWY 2L approach. The aircraft touched down 408 feet short of the runway threshold, damaging the approach lighting system and the aircraft. The captain disconnected the autopilot at about 200 feet AGL.
- Categorization: Near Miss
- Reasoning: While there was substantial damage to the aircraft and the approach lighting system, this incident could have been more severe if not for the captain's actions. The aircraft touched down short of the runway threshold, making it a near miss.

In summary, Event 1 is categorized as an accident due to injuries. Event 2 is a minor event, and Event 3 is a near miss due to the potential for a more serious accident during the landing.

Figure 4 - ChatGPT's Categorization of the Events

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BIOGRAPHY



Robin L. Dillon is a Professor in the McDonough School of Business at Georgetown University. Professor Dillon seeks to understand and explain how and why people make the decisions that they do under conditions of uncertainty and risk. This research specifically examines critical decisions that people have made following near-miss events in situations with severe outcomes including hurricane evacuation, terrorism, cybersecurity, and NASA mission management. She has received research funding from the National Science Foundation, NASA, the Department of Defense, and the Department of Homeland Security. She participated as a NASA Summer Faculty Fellow at Goddard Space Flight Center in 2004.



Peter M. Madsen is a professor of Organizational Behavior and Human Resources at BYU's Marriott School of Business. His research focuses on employee health and safety, organizational change management, and organizational learning from accidents, incidents, and near misses. Much of his work centers on how organizations can draw lessons from accidents and near accidents and enact change based on those lessons to improve subsequent employee safety. He received a PhD in Organizational Behavior and Industrial Relations from the Haas School of Business at the University of California Berkeley.



Brian Holland has a BS in Wireless Engineering (2005), and an MS in Electrical Engineering (2007), both from Auburn University. He worked as an RF and Electrical Engineer at Lockheed Martin, and is currently pursuing an MS in Data Science for Public Policy at Georgetown (2024). He has experience in marine and electrical engineering, stealth radar receivers, non-profit groups, and doing hands-on research.



Danniell Cao is an undergraduate student at the College of William and Mary. He graduated from Thomas Jefferson High School for Science and Technology in 2023, where he primarily took courses focused on engineering. He hopes to eventually study and work in Industrial Engineering or a similar field.